

HYDRONEXIS - AI · ACCOUNTING MODULE · PURCHASING SUB-MODULE

ACC - PUR - 002

Purchasing Mission Control

Operational control center — 5-second comprehension, no scroll

MODULE	HydroNexis-AI · Accounting · Purchasing · Screen 001
DOCUMENT ID	ACC - PUR - 002
STATUS	Ready for Development
VERSION	1.0
REPOSITORY	eyalbenari99-star/hydropro
BRANCH	claude/accounting-purchasing-suppliers-5p94sz
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Purpose

This is the first screen displayed when a user enters Accounting → Purchasing. It is the operational control center for the Purchasing Department. A trained user must comprehend the entire purchasing operation within **5 seconds**. No vertical scroll on a 27" monitor. Critical information only.

Access matrix

ROLE	ACCESS
Purchasing Officer	Full
Purchasing Manager	Full
Accounting Manager	Full
Finance Manager	Read only
CEO	Read only
Warehouse	Limited (deliveries + receiving alerts only)
Department Requester	No access

Page chrome

Header (80 px)

Left-to-right: Company logo · breadcrumb (Accounting → Purchasing → Mission Control) · current user · current date · notifications bell · AI status indicator · global search · profile menu.

Left navigation

Vertical rail. Icon + label. Sections:

- Dashboard
- Purchase Requests
- Suppliers
- RFQs
- Quotations
- Negotiations
- Purchase Orders
- Receiving
- Reports
- Administration

Main dashboard layout

12-column responsive grid. Three rows. Nothing below the fold on a 27" monitor at 100% zoom.

Row 1 — KPI cards (5)

#	CARD	DISPLAYS	CLICK TARGET
1	Purchase Requests	New · Pending · Approved · Rejected · Urgent	Opens screen 1.x with the matching status filter applied
2	RFQs	Open · Sent · Waiting · Expired	Opens 7.4 RFQ Tracking, pre-filtered
3	Supplier Responses	Unread · Today · Late · Urgent	Opens 7.5 Supplier Responses
4	Purchase Orders	Draft · Approved · Issued · Receiving · Closed	Opens 13.x
5	Budget Status	Monthly Budget · Committed · Available · Forecast	Opens 16.4 Budget vs Actual

Row 2 — Priority queue + activity timeline

Left panel · AI Priority Queue

Top 20 purchasing tasks requiring attention, ranked by Nexi. Columns: Priority · Request Number · Supplier · Issue · Due Date · AI Recommendation. Row actions: Review · Approve · Assign.

Right panel · Purchasing Activity Timeline

Reverse-chronological. Icons per event type: Created Request · Supplier Replied · PO Approved · Shipment Delayed · Goods Received · Invoice Received · Supplier Warning. Every item click-through to the source record.

Row 3 — Performance · budget · alerts

PANEL	CONTENTS
Supplier Performance (left)	Top suppliers · Average delivery · Average quality · Response time · Risk score · Trend (last 30 / 90 days)
Budget Utilization (middle)	Monthly spend · Department spend · Project spend · Budget remaining · Graph (stacked bar per month)
Alerts (right)	Late supplier · Budget exceeded · No supplier found · Duplicate request · Approval waiting · Delivery delay · Critical inventory

Floating AI panel (Nexi)

Always visible, bottom-right. Persistent across navigation inside Purchasing.

Functions:

- Find supplier
- Compare quotes
- Create RFQ
- Generate PO draft
- Summarize supplier emails

- Risk analysis
- Open chat

Universal search

Single input, keyboard shortcut  /. Searchable entities:

- Item
- Supplier
- Purchase Request
- Purchase Order
- RFQ
- Invoice reference
- SOS item code
- Barcode
- QR

Filters (dashboard-wide)

- Project
- Department
- Supplier
- Category
- Priority
- Status
- Warehouse
- Date range
- Budget code

Filters persist per user session and are echoed in the URL for shareable links.

Live updates

WebSocket subscription. Widgets update in under 2 seconds. Manual refresh is not required. If the socket drops, a "reconnecting..." chip appears and the dashboard falls back to a 15-second polling interval.

Notifications

Bell icon in the header with unread counter. Types:

- Approval required
 - Supplier reply
 - Price change
 - Inventory alert
 - Delivery delay
 - Budget alert
 - Security alert
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Colour semantics

COLOUR	MEANING
Green	Completed / on-target
Blue	Informational
Orange	Warning · attention required
Red	Critical · immediate action
Grey	Inactive / disabled

Security

- Widget-level RBAC — users only see data they are authorised for.
- No direct database access from the client.
- Every user action written to AuditLog.
- Session timeout with configurable idle window.
- MFA required for any Approve / Award action.
- All traffic over TLS; WebSocket over WSS.

APIs required

METHOD + PATH	PURPOSE
GET /api/purchasing/dashboard	Aggregated row-1 KPI card counts
GET /api/purchasing/alerts	Row-3 alerts panel
GET /api/purchasing/tasks	Row-2 AI Priority Queue
GET /api/purchasing/timeline	Row-2 activity timeline
GET /api/purchasing/kpi	Header + row-3 KPIs
GET /api/suppliers/performance	Row-3 supplier performance
GET /api/budget/status	Row-1 card 5 + row-3 budget
GET /api/sos/inventory	Live inventory hooks for alerts
GET /api/notifications	Header bell
WS /ws/purchasing	Live updates channel

Database tables used

- PurchaseRequests
- PurchaseRequestItems
- Suppliers
- SupplierPerformance
- RFQs
- RFQResponses
- PurchaseOrders
- Receiving
- Budget
- Projects
- Departments
- Users
- Notifications
- AuditLog

KPIs displayed

KPI	DEFINITION
Average Purchase Cycle	Days from PR submission to PO closed
Average Supplier Response	Median time between RFQ sent and first quote received
Savings	(Baseline price – awarded price) × qty, monthly rolling
Open RFQs	Count of RFQs whose status is Sent or Waiting
Late Deliveries	Count of PO lines past promised delivery date and not fully received
Budget Utilization	Committed + spent as % of allocated budget, monthly
Supplier Score	Weighted mean of delivery, quality, service, price ratings
Purchase Lead Time	Days from PO issued to first receipt
Inventory Risk	SKUs projected to stock-out within lead time × unit cost
Approval Time	Median hours from PR submitted to first approval action

Acceptance criteria

- Dashboard loads in under 3 seconds on the reference workstation (Chromium latest, 100 Mbps, warm cache).
- All widgets respect role-based permissions; hidden data never reaches the client.
- Real-time updates occur without page refresh within 2 seconds of the source event.
- All counts reconcile exactly with the underlying transaction data (nightly reconciliation job).
- Every alert links directly to the transaction that raised it.
- Every user action is written to AuditLog with actor, timestamp, before/after state, IP, device.
- Dashboard is responsive from 1366×768 desktop up to 4K displays; no horizontal scroll at any size.
- MFA prompt appears before any Approve / Award action from within the dashboard.

